

Networking Assignment Week 1

1. Consider a case, a folder has multiple files and how would copy it to destination machine path.

- Created a folder test_src and added files inside
- CMD USED :
 - **cp -r test_src test_dest/**
- -r is used to copy all files in folder
- Verified using ls command
- Also tested copying to another user using:
 - **scp -r test_src user2@localhost:/home/user2/**

```
boss@boss:~/Documents$ pwd  
/home/boss/Documents
```

```
boss@boss:~/Documents$ mkdir test_dest  
boss@boss:~/Documents$ cp -r test_src test_dest/  
boss@boss:~/Documents$ ls test_dest/  
test_src  
boss@boss:~/Documents$ ls test_dest/test_src  
file1.txt  file2.txt  file3.txt
```

```
boss@boss:~/Documents$ scp -r test_src user2@localhost:/home/user2/  
user2@localhost's password:  
file1.txt  
file2.txt  
file3.txt
```

2. Host a FTP and SFTP server and try PUT and GET operations.

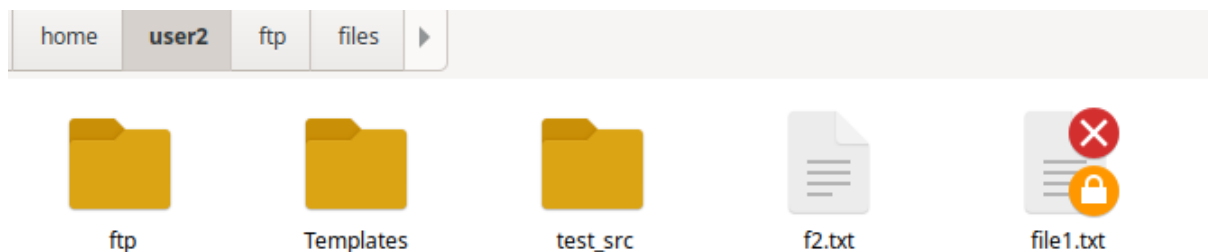
FTP:

- Installed FTP server using vsftpd on Linux
- Started FTP service using systemctl
- Found IP address using ip a
- Connected from Windows using:
 - **ftp 192.168.1.5**
- Logged in using user2 credentials
- Uploaded file using:

- **put file1.txt**
- Downloaded file using:
 - **get f2.txt**
- File transfer was successful (Transfer complete)

Server (Linux):

```
boss@boss:~$ sudo systemctl start vsftpd
sudo systemctl enable vsftpd
Synchronizing state of vsftpd.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable vsftpd
boss@boss:~$ sudo nano /etc/vsftpd.conf
boss@boss:~$ sudo systemctl restart vsftpd
boss@boss:~$ sudo chmod -R 777 /home/user2
boss@boss:~$ ip -a
```



Client (Windows):

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Manish>echo hello>file1.txt

C:\Users\Manish>ftp 192.168.1.5
Connected to 192.168.1.5.
220 (vsFTPd 3.0.3)
200 Always in UTF8 mode.
User (192.168.1.5:(none)): user2
331 Please specify the password.
Password:
230 Login successful.
ftp> put file1.txt
200 PORT command successful. Consider using PASV.
150 Ok to send data.
226 Transfer complete.
ftp: 7 bytes sent in 0.30Seconds 0.02Kbytes/sec.
ftp> get f2.txt
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for f2.txt (9 bytes).
226 Transfer complete.
ftp: 9 bytes received in 0.00Seconds 9000.00Kbytes/sec.
ftp>
```

f2	22-03-2026 10:05	Text Document	1 KB
file1	22-03-2026 10:01	Text Document	1 KB

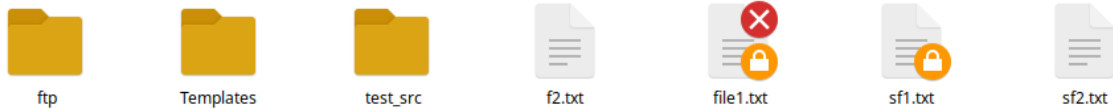
SFTP:

- Installed SSH server using openssh-server
- Started SSH service
- Connected from Windows using:
 - **sftp user2@192.168.1.5**
- Logged in successfully
- Uploaded file using:
 - **put sf1.txt**
- Downloaded file using:
 - **get sf2.txt**
- File transfer was successful

Server (Linux):

```
boss@boss:~$ sudo systemctl start ssh
boss@boss:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh
boss@boss:~$ sudo systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-03-22 19:17:56 IST; 2h 18min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 832 (sshd)
    Tasks: 1 (limit: 8388)
   Memory: 4.9M
      CPU: 36ms
  CGroup: /system.slice/ssh.service
          └─832 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 22 19:17:56 boss systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Mar 22 19:17:56 boss sshd[832]: Server listening on 0.0.0.0 port 22.
Mar 22 19:17:56 boss sshd[832]: Server listening on :: port 22.
Mar 22 19:17:56 boss systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```



Client (Windows):

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Manish>sftp user2@192.168.1.5
user2@192.168.1.5's password:
Connected to 192.168.1.5.
sftp> put sf1.txt
Uploading sf1.txt to /home/user2/sf1.txt
sf1.txt                                     100%   5    0.0KB/s   00:00
sftp> get sf2.txt
Fetching /home/user2/sf2.txt to sf2.txt
sf2.txt                                     100%   6    0.4KB/s   00:00
sftp>
```

File	Date/Time	Type	Size
sf1	22-03-2026 10:11	Text Document	1 KB
sf2	22-03-2026 10:14	Text Document	1 KB

3. Explore with Wireshark/TCP-dump/cisco packet tracer tools and learn about packet filters.

- Opened Wireshark and selected network interface
- Captured live network traffic
- Generated traffic by browsing websites
- Applied packet filters to analyze traffic

Filters used:

- tcp
- ip.addr == 192.168.1.5
- tcp.port == 443

Tcp:

No.	Time	Source	Destination	Protocol	Length	Info
18	0.489955774	2401:4900:1ce0:1424...	2404:6800:4007:82e...	TCP	86	32884 → 443 [ACK] Seq=4286 Ack=1500 Win=541 Len=0 TSval=2983388078 TSecr=1199332427
21	1.791770408	2401:4900:1ce0:1424...	2606:4700:9003:c0c9...	TCP	86	58182 → 443 [ACK] Seq=1 Ack=1 Win=631 Len=0 TSval=1790890615 TSecr=3445902244
22	3.071809155	2401:4900:1ce0:1424...	2401:4900:1ce0:1424...	TCP	86	105 → 443 [ACK] Seq=1 Ack=1 Win=631 Len=0 TSval=1790890615 TSecr=3445902244
23	3.839174340	2401:4900:1ce0:1424...	2404:6800:4007:82e...	TCP	86	45374 → 443 [ACK] Seq=1 Ack=1 Win=641 Len=0 TSval=1321443748 TSecr=329898962
24	3.891633435	2404:6800:4007:82e...	2401:4900:1ce0:1424...	TCP	86	[TCP ACKed unseen segment] 443 → 45374 [ACK] Seq=1 Ack=2 Win=271 Len=0 TSval=329944014 TSecr=13210355
25	5.059813061	35.82.117.226	192.168.1.5	TLSv1.2	185	Application Data
26	5.059825555	35.82.117.226	192.168.1.5	TLSv1.2	90	Application Data
27	5.059825705	35.82.117.226	192.168.1.5	TCP	66	443 → 42070 [FIN, ACK] Seq=64 Ack=1 Win=197 Len=0 TSval=2745191385 TSecr=797838485
28	5.059973111	192.168.1.5	35.82.117.226	TCP	66	42070 → 443 [ACK] Seq=1 Ack=65 Win=595 Len=0 TSval=797847481 TSecr=2745191385
29	5.060210495	192.168.1.5	35.82.117.226	TCP	66	42070 → 443 [FIN, ACK] Seq=1 Ack=65 Win=595 Len=0 TSval=797847482 TSecr=2745191385
30	6.052363200	35.82.117.226	192.168.1.5	TCP	66	443 → 42070 [ACK] Seq=65 Ack=2 Win=197 Len=0 TSval=2745191654 TSecr=797847482
31	6.052374602	35.82.117.226	192.168.1.5	TLSv1.2	185	Application Data
32	6.052374842	35.82.117.226	192.168.1.5	TLSv1.2	90	Application Data
33	6.052375012	35.82.117.226	192.168.1.5	TCP	66	443 → 42082 [FIN, ACK] Seq=64 Ack=1 Win=121 Len=0 TSval=2745191682 TSecr=797838537
34	6.095170521	192.168.1.5	35.82.117.226	TCP	66	42082 → 443 [ACK] Seq=1 Ack=65 Win=574 Len=0 TSval=797848517 TSecr=2745191682
35	6.266393370	192.168.1.5	9.9.9.9	TCP	74	37364 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=202415827 TSecr=0 WS=128
36	7.071542680	9.9.9.9	192.168.1.5	TCP	74	443 → 37364 [SYN, ACK] Seq=0 Ack=1 Win=43440 Len=0 MSS=1460 SACK_PERM TSval=2294314462 TSecr=20241582
37	7.071624924	192.168.1.5	9.9.9.9	TCP	66	37364 → 443 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=202416632 TSecr=2294314462

ip


```
boss@boss:~$ sudo tcpdump -i wlo1 tcp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on wlo1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
21:45:28.730011 IP6 boss.42254 > dns.google.https: Flags [P.], seq 4131836751:4131836775, ack 3122720278, win 529, options [nop,nop,TS val 2845841725 ecr 3233758520], length 24
21:45:28.730069 IP6 boss.42254 > dns.google.https: Flags [F.], seq 24, ack 1, win 529, options [nop,nop,TS val 2845841725 ecr 3233758520], length 0
21:45:28.781299 IP6 dns.google.https > boss.42254: Flags [F.], seq 1, ack 24, win 1052, options [nop,nop,TS val 3233762572 ecr 2845841725], length 0
21:45:28.781350 IP6 boss.42254 > dns.google.https: Flags [.], ack 2, win 529, options [nop,nop,TS val 2845841776 ecr 3233762572], length 0
21:45:28.785172 IP6 dns.google.https > boss.42254: Flags [.], ack 25, win 1052, options [nop,nop,TS val 3233762577 ecr 2845841725], length 0
21:45:40.131860 IP6 boss.38184 > maa05s10-in-x0a.1e100.net.https: Flags [P.], seq 2804903295:2804903334, ack 3086886103, win 604, options [nop,nop,TS val 2531989461 ecr 3444454082], length 39
21:45:40.177025 IP6 maa05s10-in-x0a.1e100.net.https > boss.38184: Flags [.], ack 39, win 968, options [nop,nop,TS val 3444512781 ecr 2531989461], length 0
21:45:40.177027 IP6 maa05s10-in-x0a.1e100.net.https > boss.38184: Flags [P.], seq 1:40, ack 39, win 968, options [nop,nop,TS val 3444512781 ecr 2531989461], length 39
21:45:40.220927 IP6 boss.38184 > maa05s10-in-x0a.1e100.net.https: Flags [.], ack 40, win 604, options [nop,nop,TS val 2531989550 ecr 3444512781], length 0
21:45:44.972443 IP boss.42312 > dns9.quad9.net.https: Flags [S], seq 3358181397, win 64240, options [mss 1460,sackOK,TS val 3341426237 ecr 0,nop,wscale 7], length 0
21:45:44.972917 IP6 boss.38184 > maa05s10-in-x0a.1e100.net.https: Flags [P.], seq 39:339, ack 40, win 604, options [nop,nop,TS val 2531994302 ecr 3444512781], length 300
21:45:45.027156 IP6 maa05s10-in-x0a.1e100.net.https > boss.38184: Flags [.], ack 339, win 967, options [nop,nop,TS val 3444517632 ecr 2531994302], length 0
21:45:45.041104 IP dns9.quad9.net.https > boss.42312: Flags [S.], seq 2334579615, ack 3358181398, win 43440, options [mss 1300,sackOK,TS val 2331885671 ecr 3341426237,nop,wscale 11], length 0
21:45:45.041160 IP boss.42312 > dns9.quad9.net.https: Flags [.], ack 1, win 502, options [nop,nop,TS val 3341426306 ecr 2331885671], length 0
21:45:45.050428 IP boss.42312 > dns9.quad9.net.https: Flags [P.], seq 1:518, ack 1, win 502, options [nop,nop,TS val 3341426315 ecr 2331885671], length 517
21:45:45.056525 IP6 maa05s10-in-x0a.1e100.net.https > boss.38184: Flags [P.], seq 40:112, ack 339, win 967, options [nop,nop,TS val 3444517662 ecr 2531994302], length 72
21:45:45.056527 IP6 maa05s10-in-x0a.1e100.net.https > boss.38184: Flags [P.], seq 112:143, ack 339, win 967, options [nop,nop,TS val 3444517662 ecr 2531994302], length 31
```

Ip:

```
boss@boss:~$ sudo tcpdump -i wlo1 host 10.236.148.182
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on wlo1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
21:53:37.113093 IP boss.46250 > _gateway.domain: 65366+ A? dns.google. (28)
21:53:37.113107 IP boss.46250 > _gateway.domain: 6482+ AAAA? dns.google. (28)
21:53:37.146334 IP boss.45089 > _gateway.domain: 44935+ PTR? 23.148.236.10.in-addr.arpa. (44)
21:53:37.251975 IP _gateway.domain > boss.46250: 65366 2/0/0 A 8.8.8.8, A 8.8.4.4 (60)
21:53:37.252080 IP _gateway.domain > boss.46250: 6482 2/0/0 AAAA 2001:4860:4860::8888, AAAA 2001:4860:4860::8844 (84)
21:53:37.252128 IP _gateway.domain > boss.45089: 44935 NXDomain 0/0/0 (44)
21:53:37.253014 IP boss.33762 > _gateway.domain: 24347+ PTR? 182.148.236.10.in-addr.arpa. (45)
21:53:37.260264 IP _gateway.domain > boss.33762: 24347 NXDomain 0/0/0 (45)
21:53:39.938596 IP 93.243.107.34.bc.googleusercontent.com.https > boss.43702: Flags [F.], seq 2366312395, ack 1310759408, win 1050, options [nop,nop,TS val 1855946003 ecr 2037132906], length 0
21:53:39.938843 IP boss.43702 > 93.243.107.34.bc.googleusercontent.com.https: Flags [P.], seq 1:25, ack 1, win 529, options [nop,nop,TS val 2037432103 ecr 1855946003], length 24
21:53:39.938871 IP boss.43702 > 93.243.107.34.bc.googleusercontent.com.https: Flags [F.], seq 25, ack 1, win 529, options [nop,nop,TS val 2037432103 ecr 1855946003], length 0
21:53:39.957119 IP boss.36581 > _gateway.domain: 22702+ PTR? 93.243.107.34.in-addr.arpa. (44)
21:53:39.981055 IP _gateway.domain > boss.36581: 22702 1/0/0 PTR 93.243.107.34.bc.googleusercontent.com. (96)
21:53:39.986036 IP 93.243.107.34.bc.googleusercontent.com.https > boss.43702: Flags [R], seq 2366312396, win 0, length 0
21:53:39.986066 IP 93.243.107.34.bc.googleusercontent.com.https > boss.43702: Flags [R], seq 2366312396, win 0, length 0
21:53:42.408407 ARP, Request who-has boss tell _gateway, length 28
21:53:42.408425 ARP, Reply boss is-at 20:58:43:d9:8c:96 (oui Unknown), length 28
21:53:45.040163 IP boss.37022 > _gateway.domain: 39186+ A? dns.google. (28)
21:53:45.040181 IP boss.37022 > _gateway.domain: 22124+ AAAA? dns.google. (28)
21:53:45.041517 IP _gateway.domain > boss.37022: 39186 2/0/0 A 8.8.4.4, A 8.8.8.8 (60)
21:53:45.041691 IP _gateway.domain > boss.37022: 22124 2/0/0 AAAA 2001:4860:4860::8844, AAAA 2001:4860:4860::8888 (84)
21:53:45.343677 IP boss.46120 > 93.243.107.34.bc.googleusercontent.com.https: Flags [S], seq 2450579917, win 64240, options [mss 1460,sackOK,TS val 2037437507 ecr 0,nop,wscale 7], length 0
21:53:45.361088 IP 93.243.107.34.bc.googleusercontent.com.https > boss.46120: Flags [S.], seq 2768518129, ack 2450579918, win 65535, options [mss 1300,sackOK,TS val 1966832460 ecr 2037437507,nop,wscale 8], length 0
```

Port:


```
boss@boss:~$ sudo tcpdump -i wlo1 port 443
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on wlo1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
21:50:18.931925 IP6 lcmaaa-ao-in-x0e.1e100.net.https > boss.52614: UDP, length 194
21:50:18.932493 IP6 boss.52614 > lcmaaa-ao-in-x0e.1e100.net.https: UDP, length 32
21:50:18.939459 IP6 boss.52614 > lcmaaa-ao-in-x0e.1e100.net.https: UDP, length 1232
21:50:18.939499 IP6 boss.52614 > lcmaaa-ao-in-x0e.1e100.net.https: UDP, length 1232
21:50:18.939519 IP6 boss.52614 > lcmaaa-ao-in-x0e.1e100.net.https: UDP, length 575
21:50:18.987029 IP6 lcmaaa-ao-in-x0e.1e100.net.https > boss.52614: UDP, length 29
21:50:18.991657 IP6 lcmaaa-ao-in-x0e.1e100.net.https > boss.52614: UDP, length 35
21:50:19.011905 IP6 boss.52614 > lcmaaa-ao-in-x0e.1e100.net.https: UDP, length 33
21:50:19.415095 IP6 lcmaaa-ao-in-x0e.1e100.net.https > boss.52614: UDP, length 1015
21:50:19.415642 IP6 boss.52614 > lcmaaa-ao-in-x0e.1e100.net.https: UDP, length 39
21:50:19.430637 IP6 pnmaaa-as-in-x0a.1e100.net.https > boss.50995: UDP, length 120
21:50:19.430959 IP6 boss.50995 > pnmaaa-as-in-x0a.1e100.net.https: UDP, length 32
21:50:19.589389 IP6 lcmaaa-ao-in-x0e.1e100.net.https > boss.52614: UDP, length 29
21:50:20.051924 IP6 lcmaaa-ao-in-x0e.1e100.net.https > boss.52614: UDP, length 41
21:50:20.052339 IP6 boss.52614 > lcmaaa-ao-in-x0e.1e100.net.https: UDP, length 32
21:50:22.282196 IP6 lcmaaa-aq-in-x0a.1e100.net.https > boss.57442: UDP, length 161
21:50:22.282197 IP6 lcmaaa-aq-in-x0a.1e100.net.https > boss.57442: UDP, length 25
21:50:22.282693 IP6 boss.57442 > lcmaaa-aq-in-x0a.1e100.net.https: UDP, length 32
```

4. Understand linux utility commands like - ping, arp (Understand each params from ifconfig output)

ping:

- Used to check network connectivity
- Command used:

ping [google.com](https://www.google.com)

- Observed replies from server with time delay
- Shows packets sent, received and packet loss

```
boss@boss:~$ ping google.com
PING google.com(pnmaaa-an-in-x0e.1e100.net (2404:6800:4007:833::200e)) 56 data bytes
64 bytes from pnmaaa-an-in-x0e.1e100.net (2404:6800:4007:833::200e): icmp_seq=1 ttl=116 time=22.2 ms
64 bytes from pnmaaa-an-in-x0e.1e100.net (2404:6800:4007:833::200e): icmp_seq=2 ttl=116 time=179 ms
64 bytes from pnmaaa-an-in-x0e.1e100.net (2404:6800:4007:833::200e): icmp_seq=3 ttl=116 time=31.7 ms
64 bytes from pnmaaa-an-in-x0e.1e100.net (2404:6800:4007:833::200e): icmp_seq=4 ttl=116 time=78.4 ms
64 bytes from pnmaaa-an-in-x0e.1e100.net (2404:6800:4007:833::200e): icmp_seq=5 ttl=116 time=48.7 ms
64 bytes from pnmaaa-an-in-x0e.1e100.net (2404:6800:4007:833::200e): icmp_seq=6 ttl=116 time=49.5 ms
64 bytes from pnmaaa-an-in-x0e.1e100.net (2404:6800:4007:833::200e): icmp_seq=7 ttl=116 time=27.4 ms
64 bytes from pnmaaa-an-in-x0e.1e100.net (2404:6800:4007:833::200e): icmp_seq=8 ttl=116 time=32.2 ms
^C
--- google.com ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7005ms
rtt min/avg/max/mdev = 22.223/58.663/179.265/48.543 ms
```

arp:

- Used to find MAC address of devices in network
- Command used:

arp -a

- Displays IP address to MAC address mapping
- Example: gateway IP mapped to MAC address

```
-----  
boss@boss:~$ sudo arp  
Address                  HWtype  HWaddress      Flags Mask    Iface  
_gateway                 ether    f2:6b:f6:e7:04:b8  C           wlo1  
boss@boss:~$ sudo arp -a  
_gateway (10.236.148.23) at f2:6b:f6:e7:04:b8 [ether] on wlo1
```

ifconfig:

- Used to view network interface details
- Command used:

ifconfig

Parameters observed:

- **inet** → IP address (192.168.1.5)
- **ether** → MAC address
- **mtu** → Maximum Transmission Unit (1500)
- **RX/TX** → packets received and transmitted

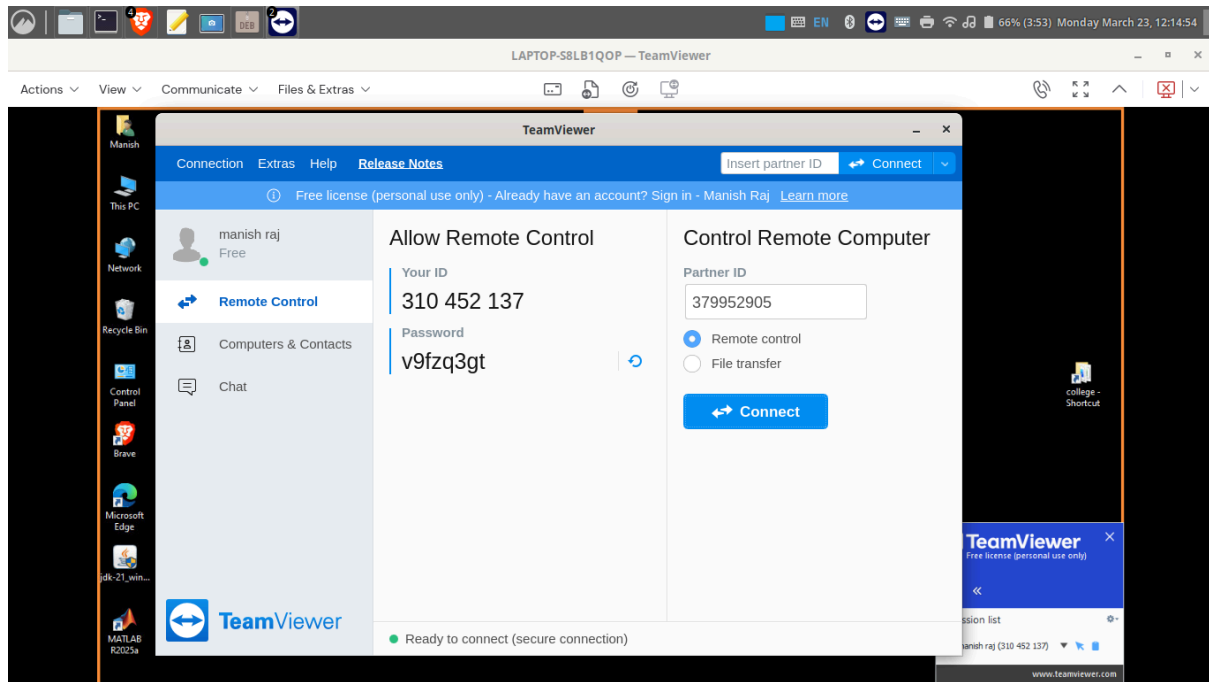

```
boss@boss:~$ sudo ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 67805 bytes 5244537 (5.0 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 67805 bytes 5244537 (5.0 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlo1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.236.148.182 netmask 255.255.255.0 broadcast 10.236.148.255
    inet6 fe80::a222:3800:9be2:3274 prefixlen 64 scopeid 0x20<link>
    inet6 2409:40f4:1038:6e6e:e385:f9b8:d7ff:8ad4 prefixlen 64 scopeid 0x0<global>
    ether 20:58:43:d9:8c:96 txqueuelen 1000 (Ethernet)
    RX packets 90316 bytes 59561881 (56.8 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 64411 bytes 30740444 (29.3 MiB)
    TX errors 0 dropped 271 overruns 0 carrier 0 collisions 0
```

5. Understand what happens when duplicate IPs are configured in a network.

- When duplicate IP addresses are configured, two or more devices share the same IP
- This causes network confusion and communication issues
 - Packet loss occurs
 - Network becomes unstable
 - ARP conflicts happen (wrong MAC mapping)
 - Devices may not connect to network properly

6. Understand how to access remote system using (VNC viewer, Anydesk, teamviewer and remote desktop connections)



- Installed TeamViewer on both systems
- Opened application and obtained ID and password
- Entered remote ID in client system
- Established connection successfully
- Accessed and controlled remote system

7. How to check if your default gateway is reachable or not and understand the default gateway.

- Default gateway is the router used to connect the local network to other networks (internet)
- Found default gateway using:

ip route

- Identified gateway IP as 192.168.1.1
- Checked reachability using:

ping 192.168.1.1

- Received replies with 0% packet loss, confirming the gateway is reachable

```
boss@boss:~$ ip route
default via 10.236.148.23 dev wlo1 proto dhcp src 10.236.148.182 metric 600
10.236.148.0/24 dev wlo1 proto kernel scope link src 10.236.148.182 metric 600
boss@boss:~$ ping 10.236.148.182
PING 10.236.148.182 (10.236.148.182) 56(84) bytes of data.
64 bytes from 10.236.148.182: icmp_seq=1 ttl=64 time=0.036 ms
64 bytes from 10.236.148.182: icmp_seq=2 ttl=64 time=0.084 ms
64 bytes from 10.236.148.182: icmp_seq=3 ttl=64 time=0.081 ms
64 bytes from 10.236.148.182: icmp_seq=4 ttl=64 time=0.074 ms
64 bytes from 10.236.148.182: icmp_seq=5 ttl=64 time=0.076 ms
64 bytes from 10.236.148.182: icmp_seq=6 ttl=64 time=0.082 ms
^C
--- 10.236.148.182 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5110ms
rtt min/avg/max/mdev = 0.036/0.072/0.084/0.016 ms
```

8. Check iwconfig/ifconfig to understand in detail about network interfaces (check about interface speed, MTU and other parameters)

- Checked network interface details using:

ifconfig
iwconfig

From ifconfig:

- **inet** → IP address (192.168.1.5)
- **ether** → MAC address (20:58:43:e6:8e:bc)
- **mtu** → Maximum Transmission Unit (1500)
- **RX/TX** → packets received and transmitted

From iwconfig:

- **ESSID** → Connected WiFi network (Airtel_siva_3012)
- **Bit Rate** → Interface speed (292.6 Mb/s)
- **Frequency** → 5.22 GHz
- **Link Quality** → signal strength

```
boss@boss:~$ sudo ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 68792 bytes 5341457 (5.0 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 68792 bytes 5341457 (5.0 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlo1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.236.148.182 netmask 255.255.255.0 broadcast 10.236.148.255
    inet6 fe80::a222:3800:9be2:3274 prefixlen 64 scopeid 0x20<link>
    inet6 2409:40f4:1038:6e6e:e385:f9b8:d7ff:8ad4 prefixlen 64 scopeid 0x0<global>
    ether 20:58:43:d9:8c:96 txqueuelen 1000 (Ethernet)
    RX packets 91818 bytes 60003319 (57.2 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 66122 bytes 31845801 (30.3 MiB)
    TX errors 0 dropped 271 overruns 0 carrier 0 collisions 0

boss@boss:~$ sudo iwconfig
lo          no wireless extensions.

wlo1       IEEE 802.11  ESSID:"Redmi Note 12 Pro+ 5G"
           Mode:Managed  Frequency:2.457 GHz  Access Point: F2:6B:F6:E7:04:B8
           Bit Rate=72.2 Mb/s   Tx-Power=20 dBm
           Retry short limit:7   RTS thr:off   Fragment thr:off
           Encryption key:off
           Power Management:on
           Link Quality=70/70  Signal level=-38 dBm
           Rx invalid nwid:0  Rx invalid crypt:0  Rx invalid frag:0
           Tx excessive retries:86  Invalid misc:19  Missed beacon:0
```

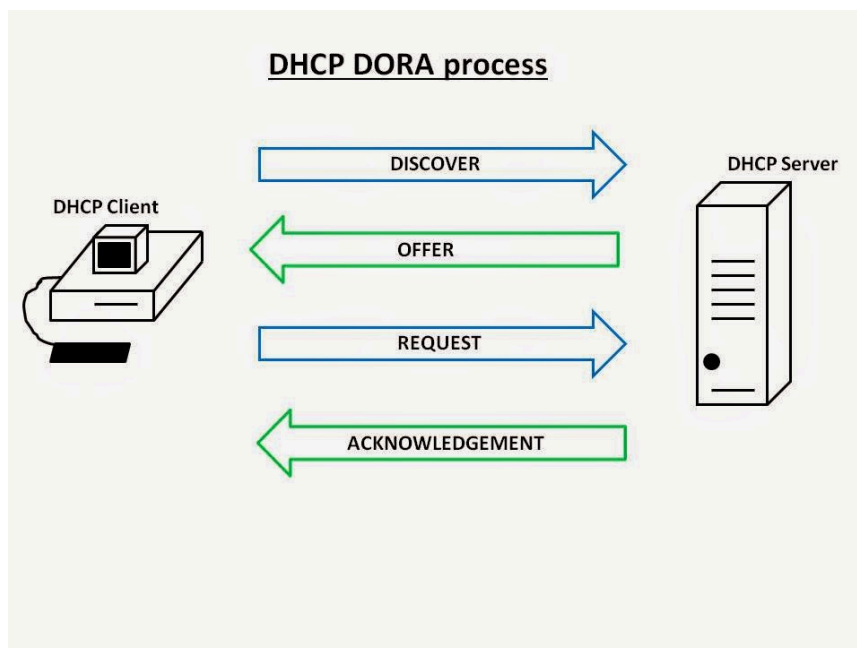
9. Log in to your home router's web interface (usually at 192.168.1.1 or 192.168.0.1) and check the connected devices list.

- Opened web browser and entered router IP address (192.168.1.1)
- Logged in using router credentials
- Navigated to "Current Wireless Users" section
- Viewed list of connected devices
- Observed details like IP address, MAC address and device name

The screenshot shows a web browser at the URL <https://192.168.1.1/status/current-wireless-user>. The page title is 'Status / Current Wireless Users'. On the left is a sidebar menu with options: Status, Device Information, Traffic Statistic, Current Wireless Users (highlighted), DHCP Lease, ARP Status, TR-069 Status, Advanced Setup, WAN Connections, LAN Setup, Dynamic DNS, Static Routing, TLS, ARP, Firewall Setup, WiFi Setup, Maintenance, and VoIP. The main content area is titled 'Current Wireless Users' and includes filters for 'WiFi Band' (2.4G) and 'SSID Index' (1). It states 'Total Clients: 1' and displays a table with the following data:

#	MAC	IP	Host Name	Receiving Rate (Mbit/s)	Sending Rate (Mbit/s)	Duration (s)	RSSI (dBm)
1	40:9F:38:63:D4:F9	192.168.1.4	LAPTOP-S8LB1QOP	150	17	9414	-67

10. Explain how a DHCP server assigns IP addresses to devices in your network.



- DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses to devices in a network

Working process (DORA):

- Device sends DHCP Discover message to find server
- DHCP server sends DHCP Offer with available IP address
- Device sends DHCP Request to accept the IP
- Server sends DHCP Acknowledgement (ACK) confirming assignment
- The device is then assigned an IP address and can communicate in the network

11. Using a terminal, connect to a remote machine via SSH and telnet.

SSH:

- Installed and started SSH service on Linux
- Connected from Windows using:

ssh user2@192.168.1.5

- Logged in using credentials
- Successfully accessed remote terminal

```
boss@boss:~$ sudo systemctl start ssh
```

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Manish>ssh user2@192.168.1.5
user2@192.168.1.5's password:
Linux boss 6.12.43+deb12-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.43-1~bpo12+1 (2025-09-06) x86_64

The programs included with the BOSS GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

BOSS GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
user2@boss:~$ pwd
/home/user2
user2@boss:~$ ls
f2.txt  file1.txt  .git  sf1.txt  sf2.txt  templates  test.cpe
user2@boss:~$
```

```
boss@boss:~$ sudo systemctl start openbsd-inetd
```

```
boss@boss:~$ sudo netstat -tulnp | grep :23
```

```
tcp    0    0 0.0.0.0:23    0.0.0.0:*    LISTEN
```

```
C:\WINDOWS\system32>dism /online /Enable-Feature /FeatureName:TelnetClient

Deployment Image Servicing and Management tool
Version: 10.0.19041.3636

Image Version: 10.0.19045.6466

Enabling feature(s)
[=====100.0%=====]
The operation completed successfully.

C:\WINDOWS\system32>telnet 192.168.1.5
```